



IMPACT OF RATIONAL FACTORS ON INVESTORS' DECISION MAKING AT INDIAN STOCK MARKET

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ABSTRACT

Decision-making is one of the critical processes which involve analysing choices that result in the most optimal level of benefit or utility for the individual. Most conventional economic theories are created and used under the assumption that all individuals taking part in an investment activity are behaving rationally. Hence the authors are interested in studying the various rational factors affecting investment decision based on their demographical levels. The main objective of the study is to explore rational investment factors and examination of the impact of these factors on investors' investment decision of Indian stock market. Data for this research was collected from a total of 1200 responses of individual investors out of which 600 from Bangalore City and 600 from Anantapur Town to draw comparison between individual investors of Bangalore City and Anantapur Town. Ten Brokerages were selected in both areas by means of multistage sampling with 60 investors under each brokerage. A Structured questionnaire with the help of Likert 5-point scale was used to record the responses of individual investors. The collected data were systematically arranged, tabulated and analyzed using appropriate statistical tools with the help of SPSS and AMOS software. Structural Equation Modelling (SEM) was used to present the Modelling between various rational constructs and investment decision. Five hypotheses were framed on the basis of rational factors that have significant positive effect on the investor's decision making. The result shows that financial parameters and project details are not influencing the decision making of the investors.

Keywords: Rational investment factors, investors' investment decisions, details in prospectus, project details and financial parameter

Introduction

Stock market is a market where stocks are bought and sold (Zuravicky, 2005, p. 6; Luu, 2011). In an economy, besides playing the role of a source for financing investment, stock market also performs a function as a signalling mechanism to managers regarding investment decisions, and a catalyst for corporate governance (Samuel, 1996, p.1). People are interested in stock because of "long-term growth of capital, dividends, and a hedge against the inflationary erosion of purchasing power" (Teweles & Bradley, 1998, p. 8). The individual investor plays a vital role in the stock market because of their good savings. The regulators of the stock market cannot ignore the behaviour of individual investors. Many individuals find investments to be fascinating because they can participate in the decision-making process and see the results of their choice. Not all investments will be profitable, as investors' whims not always result in fruitful returns. Several rational factors influence their investment decisions in stock markets.

Literature Review

Ming-Ming Lai, K. L. T. Low, and Ming-Ling Lai (2001) mentioned that the results show no significant difference in investment techniques used on bullish or bearish markets. Malaysian investors conversely seem rational and cautious about taking investment decisions. They spend a lot of time enhancing the knowledge within a firm and outside a firm until it is rendered any stock selected in the best way. Timothy C Johnson (2002) discussed that the research advances the hypothesis that some or all of the momentum effect could be responsible for stochastic growth rates. The claim works because stock prices are highly sensitive, nonlinear, and dependent on growth rates. All things equal, the recent output is associated with projected growth rate rates, which are monotonically risk-related. This relationship was demonstrated analytically through a basic partial-equilibrium model that has appeared in the literature before. Brooks & A. Katsaris (2003) mentioned whether this variability could be explained by reference to a speculative bubble. Three empirical differences Methodologies used: testing of variance limits, testing of bubble specifications, and co integration checks based on data from both ex-post and ex pre. We see why stock prices do in the last six years, they diverged dramatically from their fundamental beliefs, and that this differentiation has all the critical characteristics of a

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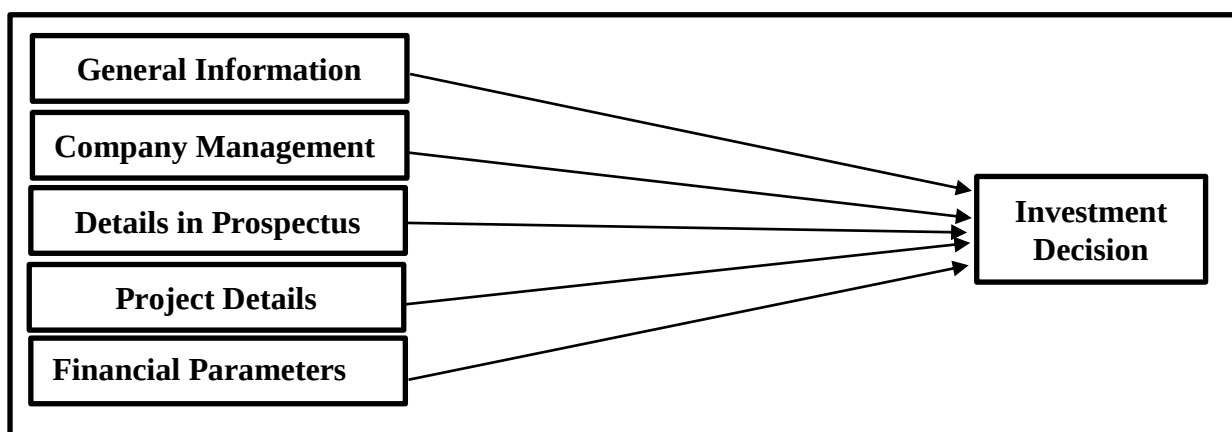
bubble. Wood and Zaichkowsky, (2004) argued that male investors are likewise more risk-tolerant than female investors. Female investors tend to be more conservative by investing more in long-term investments than male investors, relying more on brokers, and being less risk-tolerant. Besides, male investors use more internet or online trading than female investors do. Bennet and Selvam (2011) found that the market value of equity shares was affected by SPERTEL 9 (Social, Political, Economic, Regulatory, Technological, Environmental and Legal) risks. Retail investors' stock selection decision in India was affected by market factors. Huei-Wen Lin (2011) examined the relationship between rational decision making and behavioural biases. He compares the relative differences of three behavioural biases, namely the influence of disposition, herding, and overconfidence, by Specific demographic variables.

In view of this, it is necessary to establish whether rational factors influence decisions by individual investors trading at the Indian Stock Market (ISE). Whereas many studies have been carried out in other developed financial markets, little is known about the effects of rational factors on individual investors' decision making in India.

To get the research objectives, some questions are raised during the study. The study is done through answering these following questions:

- Whether there is impact of rational variables on individual investment decision making at the Indian Stock Exchange?
- At which impact levels (if any) do the rational factors influence the individual investment decisions at the Indian Stock Exchange?

Proposed Research Model



The study precisely aimed to measure the impact of rational factors on individual investor investment decision making of Indian Stock Market.

Hypotheses

H01: Rational Investment Factor General Information is significant positive influence on investment decisions of individual investors towards Indian stock market.

H02: Company Management is significant positive influence on investment decisions of individual investors towards Indian stock market.

H03: Details in Prospectus is significant positive influence on investment decisions of individual investors towards Indian stock market.

H04: Project Details is significant positive influence on investment decisions of individual investors towards Indian stock market.

H05: Financial Parameters is significant positive influence on investment decisions of individual investors towards Indian stock market.

Methodology

The primary data were collected from individual investors located in the cities Bengaluru and Anantapur by employing multi-stage sampling. A total of 1200 responses of individual investors were collected using structured questionnaire out of which 600 from Bangalore City and 600 from Anantapur Town.

Limitations:

The study is limited to rational investment factors viz., general information of the company, company management, details in prospectus, project details and financial parameters.

Measurements

The questionnaire was based on the theories of behavioral finance: Heuristic theory, Prospect theory and other theories. It consists of five parts. Part A covers demographic and economic profile of the individual investors. Part B describes perception of

investors. Part C entails social factors. Part D consists of rational investment factors and Part E involves behavioral investment factors. The 5-point Likert scale ranging from strongly agree-5, agree-4, neutral-3, disagree-2, strongly disagree-1 was used.

Data Analysis and Results

The collected data were systematically arranged, tabulated and analyzed using appropriate statistical tools with the help of SPSS and AMOS software. The structural equation modelling was carried in two distinct steps in the study. In the first step the measurement model with all the constructs was developed and later in the next step the structural model with observed (Independent) and latent (dependent) constructs was developed.

Measurement model

The measurement model with all the constructs like general information, company management, details in prospects, project details, financial parameters and investment decision was developed. Schematic presentation of the study constructs was presented in figure 2.

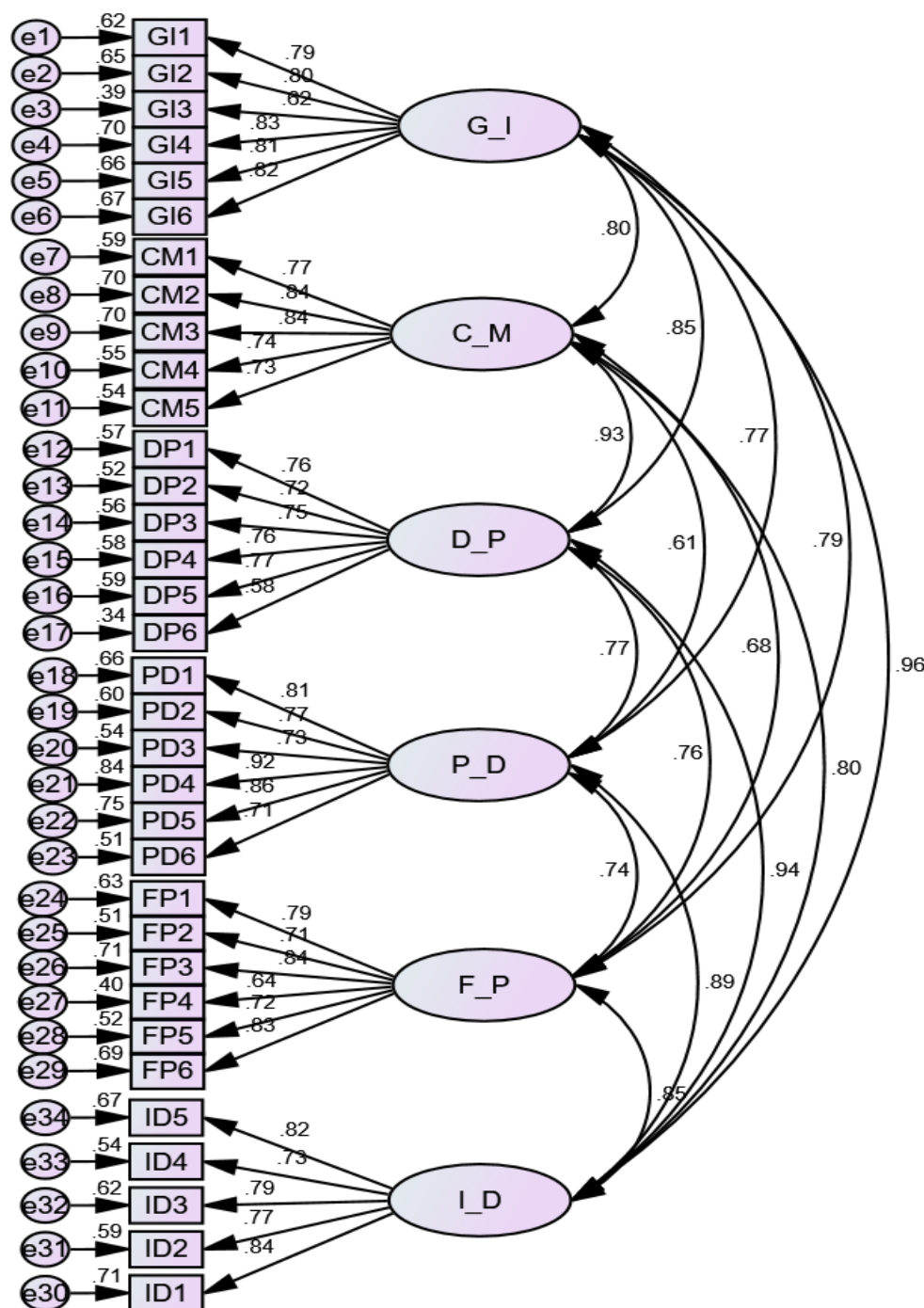


Figure.2 Measurement Model

Validity and Reliability

The measurement model fitness, reliability and validity can be assessed through three important measures namely, individual item reliabilities, convergent validity and discriminant validity (Hulland, 1999). The item reliability can be acclaimed through factor loadings. The factor loadings should be greater than 0.5. Any item below 0.5 is discarded and not considered for the analysis (Hair et al 2000). All the factor loadings for their respective constructs are greater than the standard level 0.5. So, all the items possess the reliability.

The estimation of convergent validity is carried for each construct. The Average Variance Extracted (AVE) and Composite Reliability (CR) are calculated to assess the convergent validity for each construct. The AVE and CR values if greater than 0.5 and 0.7 respectively, then it is deemed that the measurement model possesses the convergent validity. The results of the measurement model reveal that, all the factor loadings, AVE and CR values are greater than the standard levels. Hence, it is concluded that the measurement model possesses the convergent validity.

Table 1 Results of Measurement Model

Construct	Item	Factor Loadings	AVE	CR
General Information	Stock exchange information in purchasing stocks	0.790	0.613	0.904
	Considering risk factors before investing	0.804		
	Lead managers image before investment	0.622		
	Credit rating before I invest	0.835		
	Brokers advice/analyst forecast/advertisement influences me in purchasing stock	0.810		
	Price changes of stocks	0.817		
Company Management	Company history before investment	0.769	0.616	0.889
	Promoters background & contribution	0.836		
	Board of directors of company	0.839		
	Company's present policies	0.743		
	Performance of the company	0.732		
Details in Prospectus	Authorized and paid up capital of company	0.812	0.649	0.917
	Size of present issue	0.772		
	Objective of present issue	0.734		
	Terms of issue shares of company	0.919		
	Minimum and maximum subscription	0.865		
	Data of Institutional investments	0.714		
Project Details	Cost of the project and means of financing before investment	0.756	0.526	0.869
	Location/process/infrastructure of the project	0.719		
	Project and strength	0.750		
	Existing and future demand of project	0.761		
	Future projects of the company on hand	0.765		
	Future prospects and profitability of the project	0.583		
Financial Parameters	EPS/PE Ratio before investment	0.794	0.577	0.89
	Dividend payment policy	0.714		
	Book value, market value per share and price trend	0.843		
	Market volume trend	0.636		
	Historical Bonus/ right issues information	0.721		
	Performance of related companies	0.830		
Investment Decision	Investment decision based on interest payments	0.841	0.625	0.893
	Investment decision based on quick repayment of principal investment	0.771		
	Investment has lower risk compared to the market in general	0.785		
	Investment reports better results than expected	0.732		
	Investment decision made in stocks which has a high degree of safety	0.818		

Finally, the assessment of the discriminant validity is carried. The square root of AVE and inter construct correlations are compared for determining the discriminant validity. It is presumed that discriminant validity is established when the square root of AVE is greater than the inter construct correlations.

Table 2. Discriminant validity

	AVE	G_I	C_M	D_P	P_D	F_P	I_D
G_I	0.613	0.783					
C_M	0.616	0.712	0.785				
D_P	0.649	0.768	0.602	0.806			
P_D	0.526	0.708	0.717	0.715	0.725		
F_P	0.577	0.754	0.684	0.740	0.759	0.760	
I_D	0.625	0.762	0.759	0.786	0.737	0.750	0.790

Table 2 reveal that each construct's square root of AVE is greater than its inter-construct correlations. It is observed that there is discriminant validity. So, study data passed through all the conditions like factor loadings, convergent validity and discriminant validity and so proceeded for developing the structural model.

Structural Model

Structural Equation Modelling (SEM) presents the Modelling of constructs general information, company management, details in prospects, project details, financial parameters and investment decision. The pictographic representation of the modelling with constructs is presented in figure 3 And their results are shown in table 3.

Table 3 SEM Results

Measure	Estimate	Acceptable	Excellent
CMIN	640.573	--	--
DF	256	--	--
CMIN/DF	2.507	> 3	> 1
GFI	0.965	<0.90	<0.90
AGFI	0.956	<0.90	<0.90
NFI	0.966	<0.90	<0.90
RFI	0.968	<0.90	<0.90
IFI	0.950	<0.90	<0.90
TLI	0.968	<0.90	<0.90
CFI	0.967	<0.95	>0.95
SRMR	0.059	>0.08	<0.08
RMSEA	0.039	>0.06	<0.06
PClose	0.005	<0.05	>0.05

The normed chi square (CMIN/DF) value is 2.50, Root Mean Square Error Approximation (RMSEA) value 0.039 are meeting the minimum acceptable standard. The model fit indexes like Goodness of fit indices (GFI), Adjusted fit indices (AGFI), Normed fit indices (NFI), Relative fit indices (RFI), Incremental fit indices (IFI), Tucker and Lewis indices (TLI) and comparative fit indices (CFI) are meeting the minimum acceptable standard of <0.9. all these indicate that the model has good fit. The R^2 value 0.76 (Figure 3) accounts to 76 per cent of the variation in the investment decisions through five rational factors.

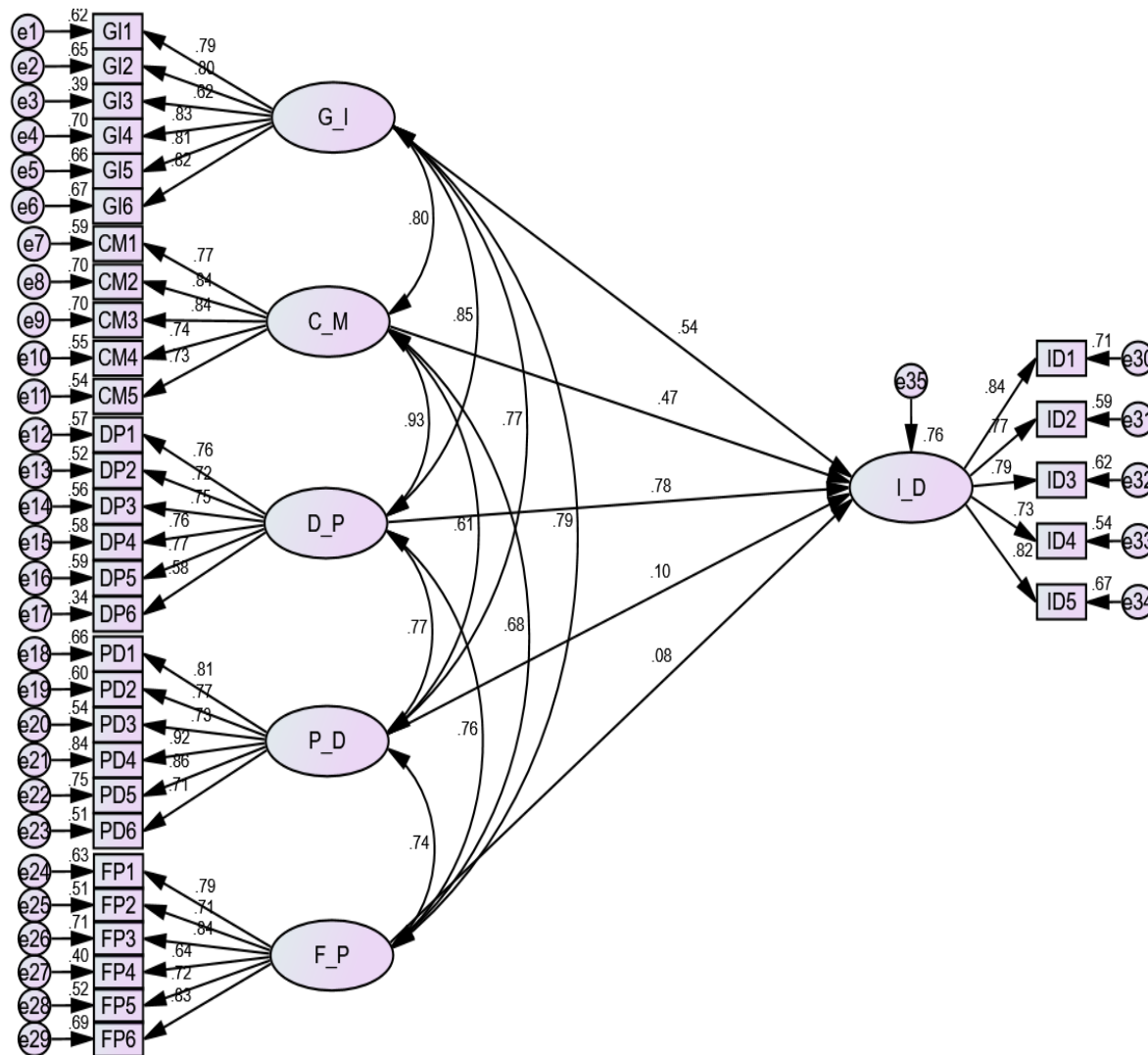


Figure 3 Structural Model

Table 4 Structural Model Summary

Hypo- Thesis	DV	IV	Beta Value	S.E.	Sig.	Result
H1	Investment Decision	<= General Information	0.541	.081	***	Supported
H2	Investment Decision	<= Company Management	0.468	.084	***	Supported
H3	Investment Decision	<= Details in Prospects	0.780	.264	***	Supported
H4	Investment Decision	<= Project Details	.102	.092	.268	Not Supported
H5	Investment Decision	<= Financial Parameters	.077	.051	.132	Not Supported

The hypotheses results and the standardized regression estimates of the structural model are depicted in table 4. The hypotheses H1, H2 and H3 are supported and H4 and H5 are not supported. The investment decision of the investor is highly influenced by the rational factor Details in prospectus (Beta Value 0.78, Significant Value 0.000). followed by general information (Beta Value 0.54, Significant Value 0.000) and Company Management (Beta Value 0.47, Significant Value 0.000).

Discussion

The only objective of the study was to identify the impact of rational factors on the investment decisions of the investor. The identified well-known rational factors from the literature which are general information, company management, details in prospectus, project details and financial parameters are considered for the study.

The hypotheses H1, H2 and H3 stated that the general information, company management and details in prospectus have significant positive effect on the investor's decision making. This shows that the general information, company management and details in prospectus expand the decision making among the investors of Indian stock market, especially, details in prospectus captured the most impact on investors decision making.

Conclusion and Scope for Further Research

The results of the study chiefly stick to previous research except with regard to project details and financial parameters. Project details and financial parameters are not influencing the decision making of the investors. The statistical insignificance of the hypothesis “H5 Financial Parameters is significant positive influence on investment decisions of individual investors towards Indian stock market” reveals a unique result. As the study was focussed on the influence of rational factors on investors’ decision making and excluded the mediating and moderating effect of demographical factors of investors in the process of investors’ decision making that paved a way for further research.

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